JEE(MAIN) : ENTHUSIAST SCORE -II

Time : 3.00 Hours **Maximum Marks : 300**

Q2

READ THE INSTRUCTIONS CAREFULLY

Important Instructions :

1. Immediately fill in the form number on this page of the Test Booklet and OMR sheet (Side-1 & Side-2) with Blue/Black Ball Point Pen. Use of pencil is strictly prohibited.
2. The candidates should not write their Form Number anywhere else (except in the specified space) on the Test Booklet/Answer Sheet.
3. The Test Booklet consists of 90 questions.
4. There are **three** parts in the question paper 1,2,3 consisting of **Physics**, **Chemistry** and **Mathematics** having **30 questions** in each subject and each subject having **Two sections**.
   (a) Section-I contains 20 **multiple choice** questions with **only one correct** option.
   Marking scheme : +4 for correct answer, 0 if not attempted and –1 in all other cases.
   (b) Section-II contains 10 **Numerical Value Type** questions. Attempt any 5 questions. First 5 attempted questions will be considered for marking.
   Marking scheme : +4 for correct answer, 0 if not attempted and –1 in all other cases.
5. No candidate is allowed to carry any textual material, printed or written, bits of papers, mobile phone any electronic device etc, except the Identity Card inside the examination hall/room.
6. Rough work is to be done on the space provided for this purpose in the Test Booklet only.
7. On completion of the test, the candidate must hand over the Answer Sheet to the invigilator on duty in the Room/Hall. **However, the candidate are allowed to take away this Test Booklet with them.**
8. **Do not fold or make any stray marks on the Answer Sheet.**
9. **Take g = 10 m/s^2 unless otherwise stated.**

Name of the Candidate (in Capitals) : _____

Form Number : in figures _____

: in words _____

Centre of Examination (in Capitals) : _____

Candidate's Signature : _____ **Invigilator's Signature :** _____

DO NOT BREAK THE SEALS WITHOUT BEING INSTRUCTED TO DO SO BY THE INVIGILATOR

Topic : Full Syllabus

PART-1 : PHYSICS

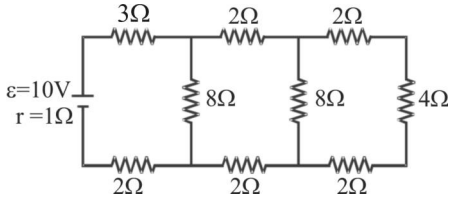
SECTION-I : (Maximum Marks: 80)

This section contains **20 questions**. Each question has 4 options for correct answer. Multiple-Choice Questions (MCQs) **Only one option is correct**. For each question, marks will be awarded as follows:

Full Marks : +4 If correct answer is selected.
Zero Marks : 0 If none of the option is selected.
Negative Marks : –1 If wrong option is selected.

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1. In the circuit shown, the cell has emf = 10V and internal resistance = 1Ω :-



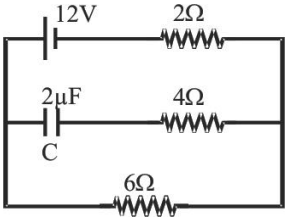
(A) The current through the 3Ω resistor is 2A

(B) The corrent through the 3Ω resistor is 0.5A

(C) The current through the 4Ω resistor is 0.5A

(D) The current through the 4Ω resistor is 0.25A

2. Find the charge on the capacitor C in the following circuit :-



(A) 12 μC (B) 14 μC

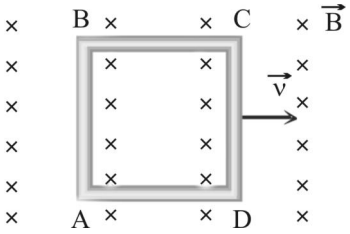
(C) 20 μC (D) 18 μC

3. A bar magnet when placed at an angle of 30° to the direction of magnetic field induction of 5×10^{-2}T, experiences a moment of couple 25×10^{-6} N-m. If the length of the magnet is 5 cm its pole strength is:-

(A) 2×10^{-2} A-m (B) 5×10^{-2} A-m

(C) 2 A-m (D) 5 A-m

4. A conducting square loop of side L and resistance R moves in its plane with a uniform velocity v perpendicular to one of its sides. A magnetic induction B constant in time and space, pointing perpendicular and into the plane of the loop exists everywhere. The current induced in the loop is :



(A) $\frac{Blv}{R}$ clockwise

(B) $\frac{Blv}{R}$ anticlockwise

(C) $\frac{2Blv}{R}$ anticlockwise

(D) Zero

5. In an ac circuit, V and I are given by

$V = 100 \sin (100\, t)$ volts,

$I = 100 \sin\left(100t + \frac{\pi}{3}\right)$ mA

The power dissipated in circuit is :

(A) 10^4 watt (B) 10 watt

(C) 2.5 watt (D) 5 watt

6. For a plane electromagnetic wave propagating in the +Z direction, which one of the following combination gives the correct possible direction for $\vec{E}$ and $\vec{B}$ field respectively ?

(A) $\left(\hat{i} + 2\hat{j}\right)$ and $\left(2\hat{i} - \hat{j}\right)$

(B) $\left(-2\hat{i} - 3\hat{j}\right)$ and $\left(3\hat{i} - 2\hat{j}\right)$

(C) $\left(2\hat{i} + 3\hat{j}\right)$ and $\left(\hat{i} + 2\hat{j}\right)$

(D) $\left(3\hat{i} + 4\hat{j}\right)$ and $\left(4\hat{i} - 3\hat{j}\right)$

7. The potential energy of a particle varies with distance x as $V = \frac{A\sqrt{x}}{x + B}$. The dimensions of AB are :-

(A) $ML^{5/2}T^{-2}$ (B) $M^1L^2T^{-2}$

(C) $M^{3/2}L^{3/2}T^{-2}$ (D) $M^1L^{7/2}T^{-2}$

8. The equation of motion of a projectile are $x = 36t$ and $2y = 96t - 9.8t^2$. Angle of projectionis :-

(A) $\text{Sin}^{-1}\left(\frac{4}{5}\right)$ (B) $\text{Sin}^{-1}\left(\frac{3}{5}\right)$

(C) $\text{Sin}^{-1}\left(\frac{4}{3}\right)$ (D) $\text{Sin}^{-1}\left(\frac{3}{4}\right)$

9. A body sliding on a smooth inclined plane requires 4 sec to reach the bottom after starting from rest at the top. How much time does it take to cover one fourth the distance starting from the top :-

(A) 1 sec (B) 2 sec

(C) 0.4 sec (D) 1.6 sec

10. A rod of length L has one end on horizontal floor. It falls while rotating about contact point without slipping. Initially it is inclined at an angle α from horizontal. Its angular speed when it is at horizontal floor is :-

(A) $\sqrt{\frac{3g \sin\alpha}{L}}$ (B) $\sqrt{\frac{2L}{3g \sin\alpha}}$

(C) $\sqrt{\frac{6g \sin\alpha}{L}}$ (D) $\sqrt{\frac{L}{g \sin\alpha}}$

11. The orbital velocity of an artificial satellite in a circular orbit very close to earth is V. The velocity of a geo-stationary satellite orbiting in circular orbit at an altitude of 3R from earth's surface will be :-

(A) $\frac{V}{\sqrt{7}}$ (B) $\frac{V}{\sqrt{6}}$

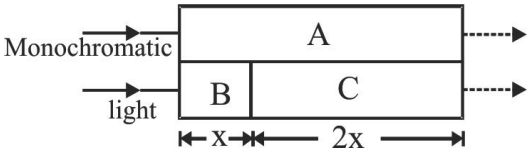
(C) $\frac{V}{2}$ (D) $\frac{V}{\sqrt{2}}$

12. One end of a uniform wire of length L and of weight W is attached rigidly to a point in the roof and a weight W_1 is suspended from its lower end. If S is the area of cross-section of the wire, the stress in the wire at a height 3L/4 from its lower end is :

(A) $\frac{W_1}{S}$ (B) $\frac{W_1 + (W/4)}{S}$

(C) $\frac{W_1 + (3W/4)}{S}$ (D) $\frac{W_1 + W}{S}$

13. Consider slabs of three media A, B and C arranged as shown in figure. R.I. of A is 1.5 and that of C is 1.4. If the number of waves in A is equal to the number of waves in the combination of B and C then refractive index of B is :-



(A) 1.4 (B) 1.5

(C) 1.6 (D) 1.7

14. White light is used to illuminate the two slits in a Young's double slit experiment. The separation between slits is b and the screen is at a distance [d >> b] from the slits. At a point on the screen directly infront of one of the slits, certain wavelengths are missing. Some of these missing wavelengths are-

[a] $\lambda = b^2/d$ [b] $\lambda = 2b^2/d$

[c] $\lambda = b^2/3\,d$ [d] $\lambda = 2b^2/3d$

(A) a, b (B) b, c (C) a, c (D) a, d

15. Cut-off potentials for a metal surface for light of wavelength λ_1, λ_2 and λ_3 are V_1, V_2 and V_3 respectively. If V_1, V_2 and V_3 are in arithmetic progression, then λ_1, λ_2 and λ_3 are in:-

(A) arithmetic progression

(B) geometric progression

(C) harmonic progression

(D) None of these

16. The binding energies per nucleon for a deutron and an α - particle are E_1 and E_2 respectively. What will be the energy E released in the following reaction ?

$_1H^2 + {}_1H^2 \longrightarrow {}_2H^4 + E$

(A) $2(E_1 + E_2)$ (B) $2(E_2 - E_1)$

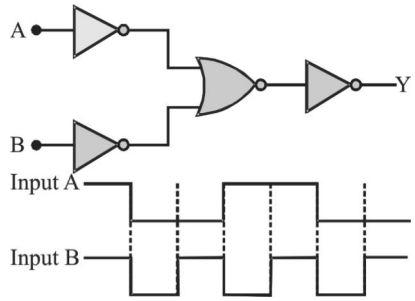
(C) $4(E_1 + E_2)$ (D) $4(E_2 - E_1)$

17. The minimum light intensity that can be perceived by human eye is about 10^{-10} Wm^{-2}. If the area of the pupil of our eye is 0.8 mm^2 and the average wavelength of visible light is 5000 Å then the number of visible light photons entering per second in our pupil will be :-

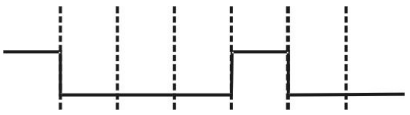
(A) 10^2 (B) 2×10^2

(C) 10^4 (D) 2×10^4

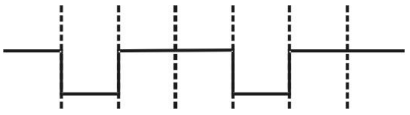
18. The logic circuit shown has the input waveforms 'A' and 'B' as shown. Pick out the **CORRECT** output waveform :-



(A)



(B)



(C)

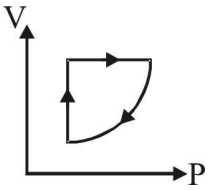


(D)

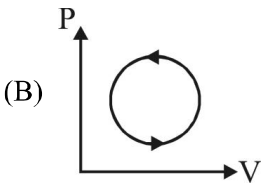


19. In the following figures heat is absorbed by the gas :-

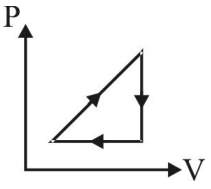
(A)



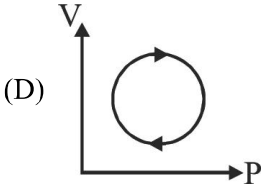
(B)



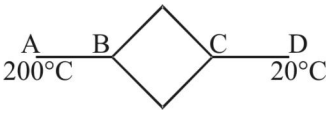
(C)



(D)



20. Six identical rods are arranged as shown in the figure. The temperature of the junction B will be :-



(A) $\frac{240}{3}^{\circ}C$ (B) 240°C

(C) 140°C (D) $\frac{140}{3}^{\circ}C$

SECTION-II : (Maximum Marks: 20)

This section contains 10 questions Candidates have to attempt any 5 questions out of 10. If more than 5 questions are attempted, then only first 5 attempted questions will be evaluated.

The answer to each question is a **Numerical Value.**

For each question, enter the correct integer value (In case of non-integer value, the answer should be rounded off to the nearest Integer).

Answer to each question will be evaluated according to the following marking scheme:

Full Marks : +4 If correct answer is entered.

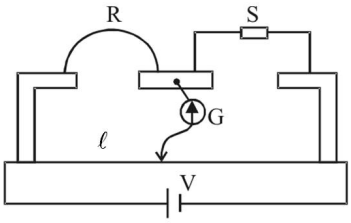
Zero Marks : 0 If the question is unanswered.

Negative Marks : –1 If wrong answer is entered.

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1. The electric potential in some region is expressed by $V = 6x - 8xy^2 - 8y + 6yz - 4z^2$ volt. The magnitude of force acting on a charge of 2 C situated at the origin will be :-

2. In a meter bridge experiment S is a standard resistance. R is a resistance wire. It Is found that balancing length is ℓ = 25 cm. If R is replaced by a wire of half length and half diameter that of R of same material, then the balancing distance ℓ' (in cm) will now be_____.



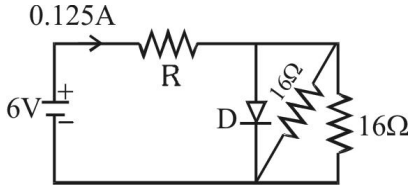
3. A particle moves in the x-y plane under the action of a force $\vec{F}$ such that the value of its linear momentum $\vec{p}$ at any time t is p_x = 2cost and p_y = 2sint. The angle θ between $\vec{F}$ and $\vec{p}$ at the time t is (in degree) :-

4. The potential energy between two atoms in a molecule is given by $U_{(x)} = \frac{16}{x^{12}} - \frac{1}{2x^6}$, where x is the distance between the atoms. The system is in equilibrium when x is equal to……

5. An object is placed on the axis of a convex lens. Its image is formed 80 cm from the object. The magnification is 3. The focal length of the lens (in cm) is :-

6. Velocity of sound in air is 330 m/s. For a particular sound in air, a path difference of 40 cm is equivalent to a phase difference of 1.6π. The frequency of the wave (in Hz) is :-

7. In the given circuit, the diode has a forward resistance of 8 Ω and infinite backward resistance. The value of R is :-



8. A nucleus of an element $_{84}X^{202}$ emits an $\alpha-$particle first, $\beta-$particle next and then a gamma photon. The final nucleus formed has an atomic number.

9. Two ideal monoatomic & diatomic gases are mixed with one another to form an ideal gas mixture. The equation of the adiabatic process of the mixture is PV^{γ} = constant, where γ = 11/7. If n_1 & n_2 are the number of moles of the monoatomic & diatomic gases in the mixture respectively, find the ratio n_1/n_2 :-

10. A black body at a temperature of 1640 K has the wavelength corresponding to maximum emission equal to 1.75μm. Assuming the moon to be a perfectly black body, the temperature of the moon, if the wavelength corresponding to maximum emission is 14.35μm is (in kelvin)

Topic : Full Syllabus

PART-2 : CHEMISTRY

SECTION-I : (Maximum Marks: 80)

This section contains **20 questions**. Each question has 4 options for correct answer. Multiple-Choice Questions (MCQs) **Only one option is correct**. For each question, marks will be awarded as follows:

Full Marks : +4 If correct answer is selected.

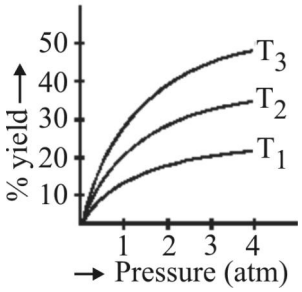
Zero Marks : 0 If none of the option is selected.

Negative Marks : –1 If wrong option is selected.

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1. The preparation of $SO_3(g)$ by reaction

$$SO_2(g) + \frac{1}{2} O_2(g) \rightleftharpoons SO_3(g)$$



is an exothermic reaction. If the preparation follows the following temperature-pressure relationship for its % yield, then for temperatures T_1, T_2 and T_3. The correct option is :-

(A) $T_3 > T_2 > T_1$

(B) $T_1 > T_2 > T_3$

(C) $T_1 = T_2 = T_3$

(D) Nothing could be predicted about temperature through given information.

2. In the following first order reactions

(A) $\xrightarrow{K_1}$ Product, (B) $\xrightarrow{K_2}$ Product , find the ratio K_1/K_2 if 90% of (A) has been reacted in time 't' while 99% of (B) has been reacted in time 2t :-

(A) 1 (B) 2

(C) 1/2 (D) $2\left(\frac{1-2\log 3}{2-\log 11-2\log 3}\right)$

3. The electrolysis of acetate solution produces ethane according to reaction

$2CH_3COO^- \longrightarrow C_2H_6(g) + 2CO_2(g) + 2e^-$

The current efficiency of the process is 80%. What volume of gases would be produced at 27°C and 740 torr, if the current of 0.5 amp is passed through the solution for 96.5 min ?

(A) 6.0 L (B) 0.60 L

(C) 1.365 L (D) 0.91 L

4. 20 ml, 0.1M HA $[(K_b)_{A^-} = 10^{-10}]$ is titrated against 0.2M, NaOH. Select the correct statement :-

[Given : log2 = 0.3, log3 = 0.48]

(A) Initially pH of HA solution is 11.5

(B) At half equivalence point pH of solution is 4 and at equivalence point pH is 9.09

(C) Phenolphthalein is suitable indicator for this titration but methyl orange is not suitable

(D) Phenolphthalein & methyl orange both are suitable indicator for end point determination.

5. If the elevation in boiling point of a solution of non volatile, non-electrolytic and non-associating solute in solvent (K_b = xK kg.mol^{-1}) is y K, then the depression in freezing point of solution of same concentration would be :
(K_f of the solvent = zK kg.mol^{-1}) -

(A) $\frac{2xz}{y}$ (B) $\frac{yz}{x}$

(C) $\frac{xz}{y}$ (D) $\frac{yz}{2x}$

6. Enthalpy of neutralization is defined as the enthalpy change when 1gm equivalent of acid/base is completely neutralized by base/acid in dilute solution. For strong acid and strong base neutralization net chemical change is
$H^+(aq) + OH^-(aq) \longrightarrow H_2O(\ell)$;
$\Delta_rH° = -55.84$kJ/mol $\Delta H°_{ionization}$ of aqueous solution of strong acid and strong base is zero.
When a dilute solution of a weak acid or based is neutralized, the enthalpy of neutralization is some what less because of the absorption of heat in the ionization of the weak acid or base, for weak acid/base
$\Delta H°_{neutralization} = \Delta H°_{ionization} + \Delta_rH°(H^+ + OH^- \longrightarrow H_2O)$
Under the same conditions how many mL of 0.1 M NaOH and 0.05 M H_2A (strong diprotic acid) solution should be mixed for a total volume of 100mL produce the highest rise in temperature :-

(A) 25 : 75 (B) 50 : 50

(C) 75 : 25 (D) 66.66 : 33.33

7. The following reaction is at equilibrium at 298K.
$2NO(0.00001 \text{ bar}) + Cl_2(g, 0.01 \text{ bar}) \rightleftharpoons 2NOCl\ (g, 0.01 \text{ bar})$
$\Delta G°$ for the reaction is :-

(A) -45.65 kJ (B) -28.53 kJ

(C) -22.83 kJ (D) -57.06 kJ

8. In which change bond order and no. of unpaired electron(s) both are increased ?

(A) $N_2 \longrightarrow N_2^+$ (B) $O_2 \longrightarrow O_2^+$

(C) $O_2^- \longrightarrow O_2^{2-}$ (D) $O_2^- \longrightarrow O_2$

9. Which of the following statements is true on the basis of back bonding ?

(A) Si–O bond is stronger than C–O bond

(B) Dimethyl ether acts a better lewis base than disilyl ether (SiH_3–O–SiH_3)

(C) $(CH_3)_3$C–O–H is less acidic than $(CH_3)_3$Si–O–H

(D) All of these

10. Which of the following having sp hybridization ?

(A) I_3^- (B) NO_3^- (C) PCl_5 (D) CO_2

11. Consider the complex ion, $[Cr(NH_3)_2Cl_2(C_2O_4)]^-$. What is the :-

(i) Oxidation number of the metal ion ?

(ii) Co-ordination number of the metal ion ?

(iii) Charge on the complex if all ligands were chloride ions ?

	(i)	(ii)	(iii)
(A)	−1	6	−3
(B)	+3	6	−3
(C)	+3	6	−1
(D)	−1	5	−3

12. In the following complex ions order of paramagnetism is :

P : $[FeF_6]^{3-}$ Q : $[CoF_6]^{3-}$

R : $[V(H_2O)_6]^{3+}$ S : $[Ti(H_2O)_6]^{3+}$

(A) P > Q > R > S (B) Q > P > R > S

(C) P = Q = R = S (D) P > R > Q > S

13. The type of isomerism present in $[Cr(NH_3)_5(NO_2)]Cl_2$ is :

(A) optical (B) linkage

(C) hydrate (D) polymerisation

14. The correct order of ionic radii of Y^{3+}, La^{3+}, Eu^{3+} and Lu^{3+} is :-

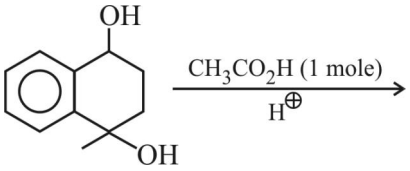
(A) $Y^{3+} < La^{3+} < Eu^{3+} < Lu^{3+}$

(B) $Y^{3+} < Lu^{3+} < Eu^{3+} < La^{3+}$

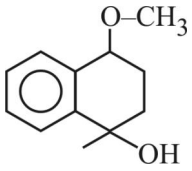
(C) $Lu^{3+} < Eu^{3+} < La^{3+} < Y^{3+}$

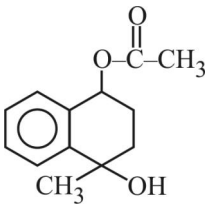
(D) $La^{3+} < Eu^{3+} < Lu^{3+} < Y^{3+}$

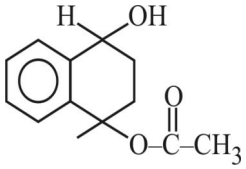
15.

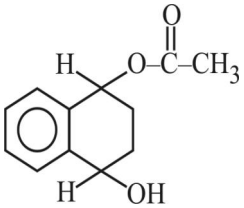


(A) , Product (A) is :

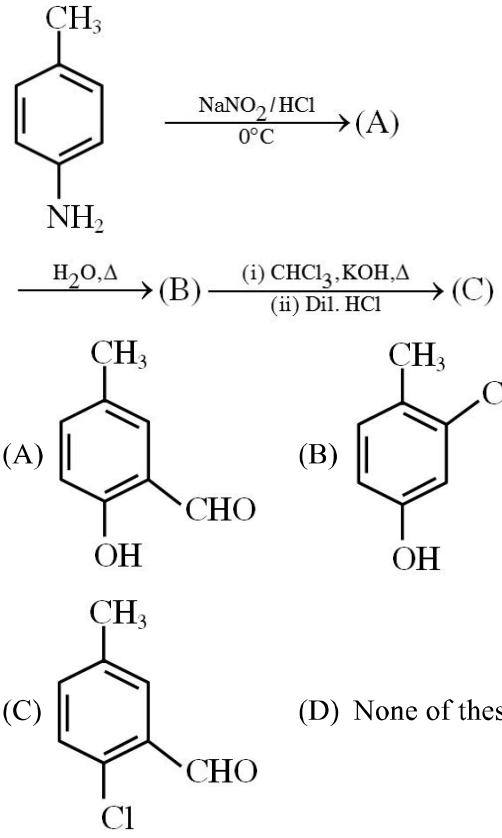
(A) 

(B) 

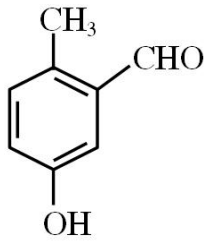
(C) 

(D) 

16. Identify C in the reactions :



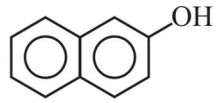
(A)

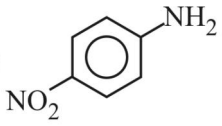
(B) 

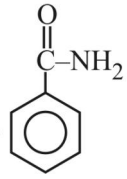
(C)

(D) None of these

17. Compound (X) reacts with $NaNO_2$ & H_2SO_4 & then reacts with ortho cresol to form sharp colour compound, Compound X may be :-

(a) 

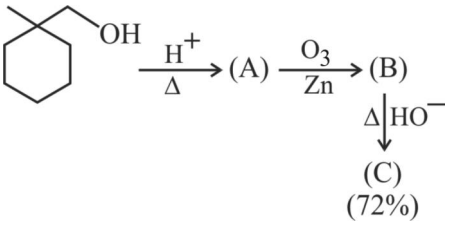
(b) 

(c) 

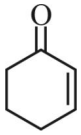
(d) 

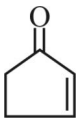
(A) a and b (B) a, b and c

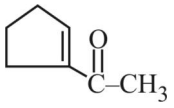
(C) b and d (D) d only

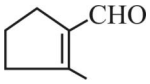
18. 

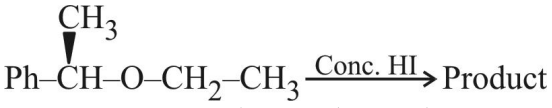
Product (C) is :-

(A) 

(B) 

(C) 

(D) 

19. 

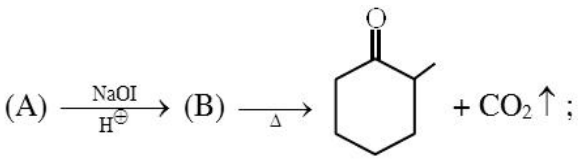
Correct statement about major product :-

(A) If follows S_N1 and racemic mixture of alcohol is obtained.

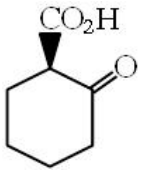
(B) It follows S_N1 and racemic mixture of alkyl iodide is obtained.

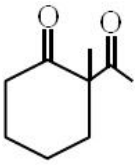
(C) It follows S_N2 and racemic mixture of alcohol is obtained.

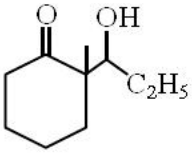
(D) If follows S_N2 and racemic mixture of alkyl iodide is obtained.

20. 

Structure of (A) is :-

(A) 

(B) 

(C) 

(D) None of these

SECTION-II : (Maximum Marks: 20)

This section contains 10 questions Candidates have to attempt any 5 questions out of 10. If more than 5 questions are attempted, then only first 5 attempted questions will be evaluated.

The answer to each question is a **Numerical Value.**

For each question, enter the correct integer value (In case of non-integer value, the answer should be rounded off to the nearest Integer).

Answer to each question will be evaluated according to the following marking scheme:

Full Marks : +4 If correct answer is entered.

Zero Marks : 0 If the question is unanswered.

Negative Marks : –1 If wrong answer is entered.

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1. For a complex reaction A $\xrightarrow{k}$ products

E_{a_1}=180 kJ/mol; E_{a_2} =80 kJ/mol; E_{a_3} = 50 kJ/mol

Overall rate constant k is related to individual rate constant by the equation $k = \left(\frac{k_1 . k_2}{k_3}\right)^{2/3}$

Activation energy (kJ/mol) for the overall reaction is :-

2. The standard electrode potential for the following reaction is +1.33V. What is the potential at pH = 2.0

$Cr_2O_7^{2-} + (aq\ 1M) + 14H^+(aq) + 6e^-$

$\longrightarrow 2Cr^{+3}(aq,1M) + 7H_2O(\ell)$

3. A solution of isopropyl alcohol and propyl alcohol has a vapour pressure 200 mm of Hg if it has 25% mole of isopropyl alcohol. Another solution of same components containing 25% mole propyl alcohol has vapour pressure 300 mm of Hg. Then vapor pressure of isopropyl alcohol in mm of Hg is:-

4. What weight of AgCl will be formed when a solution containing 4.77 g of NaCl is added to a solution of 5.77 g of $AgNO_3$?

5. Define the oxidation states of Mn in product of the given reaction

$3K_2MnO_4 + 2H_2O + 4CO_2 \longrightarrow 2X + Y + 4KHCO_3$

If the oxidation state of Mn in product X and Y are n_1 and n_2 respectively. Then find out the value of (n_1+n_2) -

6. Among the carbide how many carbide will produce methane gas on hydrolysis?

Al_4C_3, Be_2C, CaC_2, SrC_2, Mg_2C_3

7. How many of following orders are correct?

(a) Order of IE : $O^{-2} > O^+ > O > O^-$

(b) Order of EN : Zn < Cd < Hg

(c) Order of IE : B > Tl > Ga > Al > In

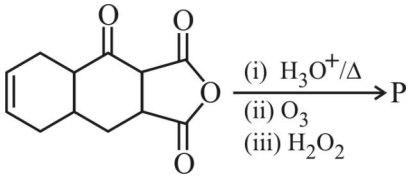
(d) Order of EA : S > Se > Te > O

(e) Order of atomic radius : Ni < Cu < Zn

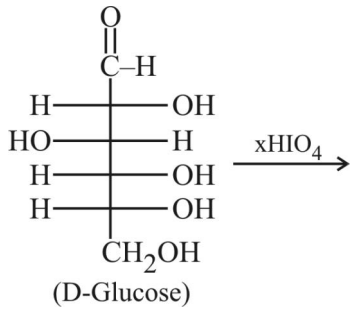
(f) Order of ionic radius : $Na^+ > Mg^{+2} > Li^+ > Be^{+2}$

(g) Order of IE_2 : N < O < F < Ne

8. The total number of carboxylic groups in product P is/are

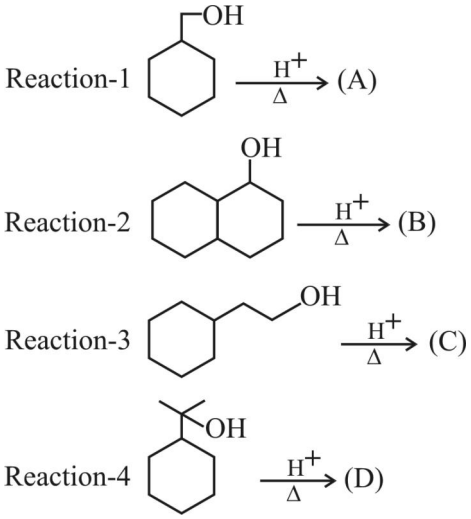


9. How many moles of HIO_4 are consumed by given compound when it react with HIO_4 :-



(D-Glucose)

10. Sum of α-hydrogen in major product of the reaction.



Sum of α-hydrogen is A + B + C + D =

Topic : Full Syllabus

PART-3 : MATHEMATICS

SECTION-I : (Maximum Marks: 80)

This section contains **20 questions**. Each question has 4 options for correct answer. Multiple-Choice Questions (MCQs) **Only one option is correct**. For each question, marks will be awarded as follows:

Full Marks : +4 If correct answer is selected.

Zero Marks : 0 If none of the option is selected.

Negative Marks : –1 If wrong option is selected.

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1. Give the correct order of initials T or F for following statements. Use T if statement is true and F if it is false.

Statement-1 : If the graphs of two linear equations in two variables are neither parallel nor identical, then there is a unique solution to the system.

Statement-2 : If the system of equations $ax + by = 0$, $cx + dy = 0$ has a non-zero solution, then it has infinitely many solutions.

Statement-3 : The system of equations $x + y + z = 1$, $x = y$, $y = 1 + z$ is inconsistent.

Statement-4 : If two of the equations in a system of three linear equations are inconsistent, then the whole system is inconsistent.

(A) FFTT (B) TTFT

(C) TTFF (D) TTTF

2. If $\alpha_1 < \alpha_2 < \alpha_3 < \alpha_4 < \alpha_5 < \alpha_6$, then the equation $(x - \alpha_1)(x - \alpha_3)(x - \alpha_5) + 3(x - \alpha_2)(x - \alpha_4)(x - \alpha_6) = 0$ has :-

(A) no real root in (α_5, α_6)

(B) no real root in (α_1, α_2)

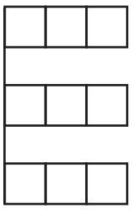
(C) all roots are imaginary

(D) no real root in $(-\infty, \alpha_1)$

3. The number of positive proper divisors of 1800, which are also divisible by 10 are

(A) 34 (B) 17 (C) 16 (D) 18

4. Number of ways in which AAABBB can be placed in the squares of figure as shown so that no row remains empty is :-



(A) 2430 (B) 2160

(C) 1620 (D) None of these

5. If two events A and B are such that $P(A^C) = 0.3$, $P(B)=0.4$ and $P(A \cap B^C)=0.5$, then value of $P\left[\frac{B}{A \cup B^C}\right]$ is equal to :-

(A) $\frac{1}{4}$ (B) $\frac{1}{2}$

(C) $\frac{3}{4}$ (D) $\frac{3}{8}$